A FINITE TYPE MODULE WITH PROPER SUPPORT IS UNIVERSALLY PURE, AND A PROPER MORPHISM GIVES UNIVERSAL PURITY

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Factorisation of a module through its scheme theoretic support. Morphisms, Section morphisms-section-support, invoked at proof line 5034; Same chapter flat.tex, label lemma-supported-on-closed, invoked at proof line 5036. Writes the module as the pushforward of a finite type module on its scheme theoretic support, so that part (1) reduces to part (2).
- A proper morphism is universally closed. Morphisms, elementary property of proper morphisms used at proof lines 5041-5042. Shows that an impurity cannot exist, since the closure of the point would have to meet the special fibre, contradicting the definition of an impurity.

Source reference. Original lemma 05K1: sources/stacks-flat/pinned-original-source.tex, source lines 4989–4989. Navigation pointer for \ref{lemma-supported-on-closed} used inside the counted span (line 5036); the argument that consumes it is retained in the span.

Source reference. Original section 056H: sources/stacks-chow/tags.txt, source lines 6710–6710. Navigation pointer for \ref{morphisms-section-support} used inside the counted span (line 5034); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes which is of finite type. Let $\mathcal{F}$ be a finite type quasi-coherent $\mathcal{O}_X$-module.*

(1) *If the support of $\mathcal{F}$ is proper over S, then $\mathcal{F}$ is universally pure relative to S.*
(2) *If f is proper, then $\mathcal{F}$ is universally pure relative to S.*

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(3) *If f is proper, then X is universally pure relative to S.*

Proof. First we reduce (1) to (2). Namely, let $Z \subset X$ be the scheme theoretic support of $\mathcal{F}$. Let $i : Z \to X$ be the corresponding closed immersion and write $\mathcal{F} = i_*\mathcal{G}$ for some finite type quasi-coherent $\mathcal{O}_Z$-module $\mathcal{G}$, see Morphisms, Section 056H. In case (1) $Z \to S$ is proper by assumption. Thus by Lemma 05K1 case (1) reduces to case (2).

Assume f is proper. Let $(g : T \to S, t' \rightsquigarrow t, \xi)$ be an impurity of $\mathcal{F}$ above $s \in S$. Since f is proper, it is universally closed. Hence $f_T : X_T \to T$ is closed. Since $f_T(\xi) = t'$ this implies that $t \in f(\overline{\{\xi\}})$ which is a contradiction. □